PhysFormer: A Physics-Guided Transformer for Physically Consistent Virtual Power Plant Forecasting

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Abstract

Virtual power plant (VPP) forecasting requires joint prediction of load, photovoltaic (PV), and wind power while avoiding non-physical outputs such as negative generation or nighttime PV power. Simple post-processing can remove some violations at deployment, but it does not shape training or explain how weather priors affect model behavior. We therefore propose PhysFormer, a physics-guided Transformer for physically consistent VPP forecasting. The model combines an explicit physical mapping stream, a bounded residual output mechanism, a physics-guided gating and alignment module, and curriculum-based constraint training. The bounded residual design yields structurally non-negative outputs under the adopted normalization, and the gating module aligns PV and wind activation with exogenous weather cues. Experiments on a real 3-year VPP dataset show that PhysFormer achieves a boundary violation rate of 0.00% and remains competitive with strong Transformer baselines in mean squared error (MSE), reaching

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0.0128 versus 0.0121 for Informer and 0.0122 for iTransformer. These results indicate that PhysFormer offers a practical balance between forecasting accuracy, physical consistency, and interpretable weather-response behavior.
Keywords: Virtual power plant, Forecasting, Physics-guided learning, Physical consistency, Transformer, Renewable power

1. Introduction

1.1. Background and Motivation

Virtual power plants (VPPs) aggregate distributed resources and coordinate them as a single operating entity. Their performance depends on accurate forecasts of load, photovoltaic (PV) generation, and wind power [1, 2, 3, 4, 5]. In practice, these forecasts must be accurate and physically plausible. PV output should vanish at night, and wind output should respect basic turbine operating limits.

Early VPP forecasting methods were mainly statistical. More recently, recurrent models and Transformers have become standard tools for multivariate time-series prediction [6, 7, 8, 9]. Recent studies have also adapted these models to load forecasting and VPP scheduling [3, 10, 11, 12]. EPSR has also reported load and wind forecasting methods that explicitly incorporate meteorological covariates and attention-based weighting of physical attributes [13, 14]. These models can reduce forecast error, but low error alone is not enough for VPP operation. Predictions should also remain consistent with basic operating conditions.

1.2. Challenges of Data-Driven VPP Forecasting

Recent advances in deep learning have improved time-series forecasting accuracy, yet their application to VPP prediction still involves several practical challenges [15, 16]:

1) Output feasibility and activation consistency: Purely data-driven models can achieve low error while still producing non-physical outputs, such as non-zero PV generation at night or negative renewable power. These violations may be small, but they reduce the value of the forecast for downstream operation.

2) Limits of post-processing and static penalties: Post-hoc clipping is useful at deployment, but it does not shape the model during training. Physics-informed penalties help, yet fixed penalties alone may not capture weather-driven activation changes in PV and wind generation [17, 18, 19]. A practical model should therefore combine train-time guidance with flexible residual prediction.

3) Need for interpretable weather-response behavior: In deployment, it is useful to know whether the model reacts to weather in a physically reasonable way. Existing interpretability tools can reveal such patterns after training, but they do not necessarily encode them into the model itself [15, 20, 16].

1.3. Overview of This Work

To address these issues, we propose **PhysFormer**, a physics-guided Transformer for physically consistent VPP forecasting. The model combines an explicit physical mapping stream, a bounded residual output mechanism, a physics-guided gating and alignment module (PGCC), and curriculum-based constraint training. In this work, PGCC is used for guidance and alignment rather than causal discovery.

The main contributions of this paper are as follows:

1. We propose a **Bounded Physical Act-Residual (BPAR)** mechanism that combines theory-based baselines with residual corrections. Under the adopted normalization, it preserves differentiability while maintaining a structurally non-negative output form.
2. We introduce a **Physics-Guided Gating and Alignment** module (PGCC) that injects weather-dependent priors into PV and wind activation. Its main role is to improve alignment and interpretability.
3. We adopt a **Physics-Constrained Curriculum Training** strategy to stabilize optimization, and we evaluate the full framework on a real 3-year VPP dataset.

Experiments show that PhysFormer eliminates boundary violations while remaining competitive with strong Transformer baselines in mean squared error. The results indicate a useful balance among accuracy, physical consistency, and engineering interpretability.

2. Proposed PhysFormer Framework

2.1. Mathematical Formulation

Throughout this paper, we adopt the following notation: P_{load}, P_{pv}, and P_{wind} denote load, PV, and wind power [MW]; G, T, and v represent irradiance [W/m^2], temperature [°C], and wind speed [m/s]; η_{pv} and β_T are the PV conversion efficiency and temperature coefficient; v_{ci}, v_r, and v_{co} are the wind turbine cut-in, rated, and cut-out speeds; a_{pv} and a_{wind} are physics activity masks $\in [0,1]$; g_{pv} and g_{wind} are PGCC gates; α is the curriculum factor; $S = 672$ and $P = 96$ are the sequence and prediction lengths; and $D = 512$ is the model dimension.

The forecasting target contains three VPP components: load (P_{load}), PV generation (P_{pv}), and wind generation (P_{wind}). Their net demand is $P_{\mathrm{net}} = P_{\mathrm{load}} - P_{\mathrm{pv}} - P_{\mathrm{wind}}$. We denote the historical component sequence by $\mathcal{X}_{\mathrm{stat}} \in \mathbb{R}^{B\times S\times 3}$, the historical weather sequence by $\mathcal{X}_{\mathrm{weather\_hist}} \in \mathbb{R}^{B\times S\times 3}$, and the future weather forecasts by $\mathcal{X}_{\mathrm{weather\_future}} \in \mathbb{R}^{B\times P\times 3}$. The model outputs $Y \in \mathbb{R}^{B\times P\times 3}$, which contains the future trajectories of load, PV, and wind.

2.2. Overall Architecture

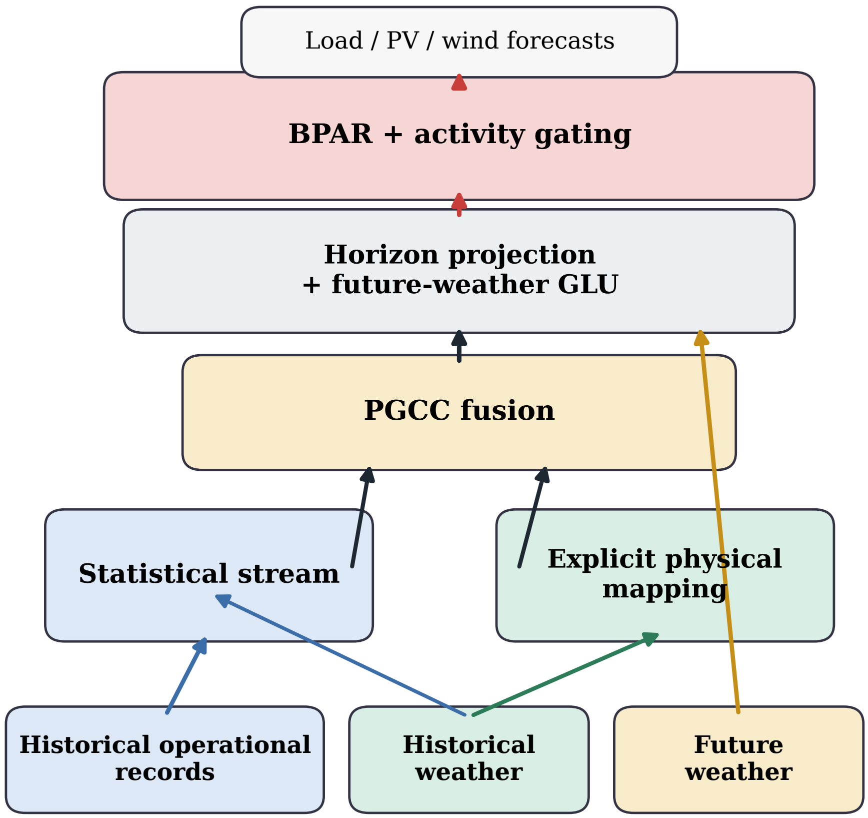

Figure 1: Overview of PhysFormer. Historical records and weather are processed by the statistical and explicit physical streams, fused by PGCC, conditioned on future weather, and converted into bounded component forecasts through BPAR with activity gating.

PhysFormer follows a compact dual-stream design. As shown in Fig. 1, historical operational records and weather are processed by a statistical stream and an explicit physical stream. The two branches are fused in PGCC, conditioned on future weather through a gated linear unit (GLU), and mapped to bounded component forecasts by BPAR with activity gating.

The statistical stream uses a Transformer encoder ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$, 8 heads, 3 layers). The physical stream produces theory-based baselines from weather variables before fusion.

2.3. Explicit Physical Mapping

The explicit physical mapping layer injects physics by mapping weather variables to theory-based power baselines.

2.3.1. PV Model

The theoretical PV generation is driven by irradiance G and degraded by cell temperature, modeled as:

$$P_{\text{pv}} = G \cdot \eta_{\text{pv}} \cdot \text{clamp}(1 - \beta_T \cdot (T - 25), 0, 1.5) \tag{1}$$

where T is the temperature. The learnable conversion efficiency is $\eta_{\text{pv}} = 0.33 \cdot \sigma(\theta_{\text{eff}})$, and the temperature coefficient is $\beta_T = \text{softplus}(\theta_{\text{temp\_coef}}) \times 0.01 \in [0.003, 0.005]/°\text{C}$.

2.3.2. Wind Model

Wind power is largely determined by three operating thresholds: the cut-in speed v_{ci}, rated speed v_{r}, and cut-out speed v_{co}. To preserve their order during training, we use a cumulative delta form: $v_{\text{ci}} = \text{softplus}(\theta_{\text{ci}})$, $v_{\text{r}} = v_{\text{ci}} + \text{softplus}(\theta_{\text{r\_delta}})$, and $v_{\text{co}} = v_{\text{r}} + \text{softplus}(\theta_{\text{co\_delta}})$. The power output then follows an idealized soft curve:

$$P_{\text{wind}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot I_{\text{running}} \tag{2}$$

where w_{norm} is the normalized speed, $scale_{\text{wind}}$ is a learnable amplitude factor, and I_{running} is 1 within $[v_{\text{ci}}, v_{\text{co}}]$ and 0 otherwise. To maintain differentiability, PhysFormer replaces I_{running} with a continuous physics activity mask a_{wind} (detailed in Section 2.3.4):

$$P_{\text{wind,actual}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot a_{\text{wind}} \tag{3}$$

This design keeps the threshold structure differentiable and avoids Straight-Through Estimators. The cumulative delta form also preserves the ordering of the operating limits during training.

2.3.3. Load Model

We abstract the power load variations triggered by human cooling and heating activities through a quadratic thermal comfort model:

$$P_{\text{load}} = P_{\text{base}} + \text{softplus}(\gamma) \cdot (T - T_c)^2 \tag{4}$$

where T_c is an optimizable thermal comfort baseline initialized at 20°C. In our design, the Transformer captures the main temporal patterns of load, while the physical mapping stream captures the temperature-driven deviation around that baseline.

Fig. 2 shows the mappings that determine the physical activation boundaries most directly. We visualize PV and wind because boundary handling is most critical in these channels. The learned load comfort parameters are reported in Table 5.

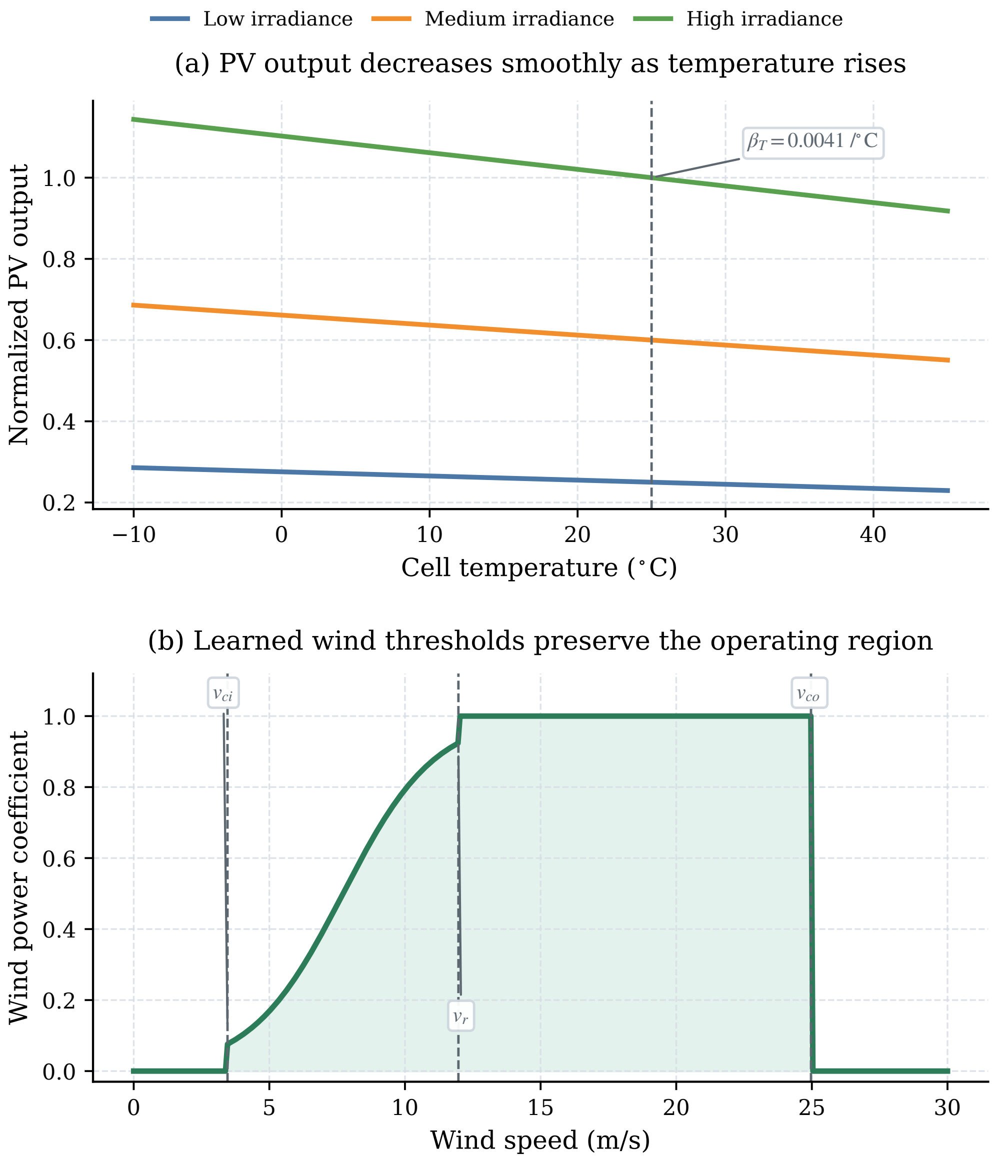

Figure 2: The learned PV and wind mappings preserve physically meaningful monotonicity and operating thresholds. The PV branch maintains smooth temperature degradation, while the wind branch retains the cut-in, rated, and cut-out structure reported in Table 5.

2.3.4. *Physics Activity Gating*

To reduce non-physical artifacts, we introduce an explicit activity gate. For PV, $a_{\text{pv}} = \tanh(10 \cdot P_{\text{pv\_theory}})$. For wind, $a_{\text{wind}} = \tanh(10 \cdot \text{softplus}(v - v_{\text{ci}}) \cdot \text{softplus}(v_{\text{co}} - v))$. The load uses $a_{\text{load}} = 1$. These masks keep the model differentiable and connect active operating states to null-generation periods.

2.4. *Physics-Guided Gating and Alignment (PGCC)*

The Physics-Guided Gating and Alignment (PGCC) module coordinates the interaction between the statistical stream and the physically encoded representations. In this work, PGCC guides weather-response behavior and improves alignment with known physical cues. It is not intended to infer causality from data alone.

2.4.1. *Physics-Anchored Illumination Prior*

We first build an illumination prior to suppress spurious activation. The prior is defined as $p_{\text{pv}}^{\text{prior}} = \sigma(k_{\text{irr}} \cdot (G_{\text{norm}} - \theta_{\text{irr}}))$, with $\theta_{\text{irr}} \in [-2.0, 0.5]$ and slope $k_{\text{irr}} \in [1, 100]$. This term separates day from night and reduces the impact of noisy activations in the data embeddings.

2.4.2. *Volatility-Aware Soft Gate*

To adapt the trust placed in the physical stream, we compute a sequence-level volatility term $\upsilon = \frac{1}{C} \sum_c \text{Var}_t(\mathcal{X}_{\text{weather},c})$. This term, together with the statistical embeddings and attention outputs, is fed to a downsampling multilayer perceptron that produces the soft gate maps.

2.4.3. Source-Differentiated Gate Bounds

The coupling implements separate activation ranges to accommodate generation source characteristics:

$$\mathrm{gate}_{\mathrm{pv}} = \mathrm{smooth}\left(p_{\mathrm{pv}}^{\mathrm{prior}} \cdot (0.5 + g_{\mathrm{soft\_pv}})\right) \in [0, 1.5] \tag{5}$$

$$\mathrm{gate}_{\mathrm{load}} = \mathrm{smooth}(g_{\mathrm{soft\_load}}) \in [0, 1.0] \tag{6}$$

The upper bound of 1.5 for $\mathrm{gate}_{\mathrm{pv}}$ leaves headroom for the residual stream. This is useful during transient conditions such as short over-irradiance events. With gate-response regularization, the learned $\mathrm{gate}_{\mathrm{pv}}$ reaches a Pearson correlation of $r = 0.8306$ with irradiance. Fig. 4 shows this alignment.

2.4.4. Physics-to-Data Curriculum Transition

We condition the alignment phase via a curriculum factor α. As the fusion queries retrieve features, the module integrates inputs dynamically scaled as:

$$q = \alpha \cdot F_{\mathrm{physics}} + (1 - \alpha) \cdot g_{\mathrm{soft}} \cdot F_{\mathrm{stat}} \tag{7}$$

At the start of training ($\alpha \to 1$), the queries rely more on the physical representations. As training proceeds ($\alpha \to 0$), the model gradually shifts toward the data-driven residual states gated by g_{soft}.

2.4.5. Temporal Smoothing

Raw gate values can be locally discontinuous. We therefore smooth all temporal gating variables with a grouped one-dimensional depthwise convolution (kernel size 3, groups=d_{model}).

2.5. Exogenous Weather Integration

After horizon projection, the model incorporates future weather information that is not available in the history sequence alone. We concatenate the future physical latent space $F_{\text{weather\_future}}$ with the history representation F_{history} and fuse them with a Gated Linear Unit:

$$F_{\text{future}} = \text{gate} \cdot \text{proj}(F_{\text{cat}}) + F_{\text{history}} \tag{8}$$

This step conditions the residual heads on upcoming weather changes such as dawn transitions or wind cut-outs.

2.6. Bounded Physical Act-Residual (BPAR) Mechanism

To reduce non-physical anomalies while preserving gradient flow, we propose the Bounded Physical Act-Residual (BPAR) mechanism. Let μ_x and σ_x denote the global mean and standard deviation of component x. We define the normalized physical zero-point as $P_{x,\text{zero}} = (0 - \mu_x)/\sigma_x$. BPAR combines the theory baseline with the neural residual through a differentiable Softplus form:

$$\hat{P}_{x,\text{raw}} = \text{Softplus}\Big((P_{x,\text{theory}} + r_x) - P_{x,\text{zero}}\Big) + P_{x,\text{zero}} \tag{9}$$

Because $\text{Softplus}(z) > 0$ for all z, this form keeps $\hat{P}_{x,\text{raw}} > P_{x,\text{zero}}$ in the normalized domain. Under the adopted normalization and anti-normalization setting, the output remains structurally non-negative for $\sigma_x > 0$. We further use a small numerical floor $\epsilon = 10^{-5}$ for degenerate variance cases such as night-time PV.

The prediction is then fused with the physics activity mask $a_x \in [0, 1]$:

$$\hat{P}_{x,\text{final}} = a_x \cdot \hat{P}_{x,\text{raw}} + (1 - a_x) \cdot P_{x,\text{zero}} \tag{10}$$

When $a_x \to 0$, BPAR smoothly pulls the output toward the physical zero baseline. Compared with unconstrained predictors, it reduces boundary violations at the cost of small local corrections near the boundary.

Because the main benefit of BPAR appears near the zero boundary, Fig. 3 visualizes the output manifold and the role of activity gating around shutdown transitions.

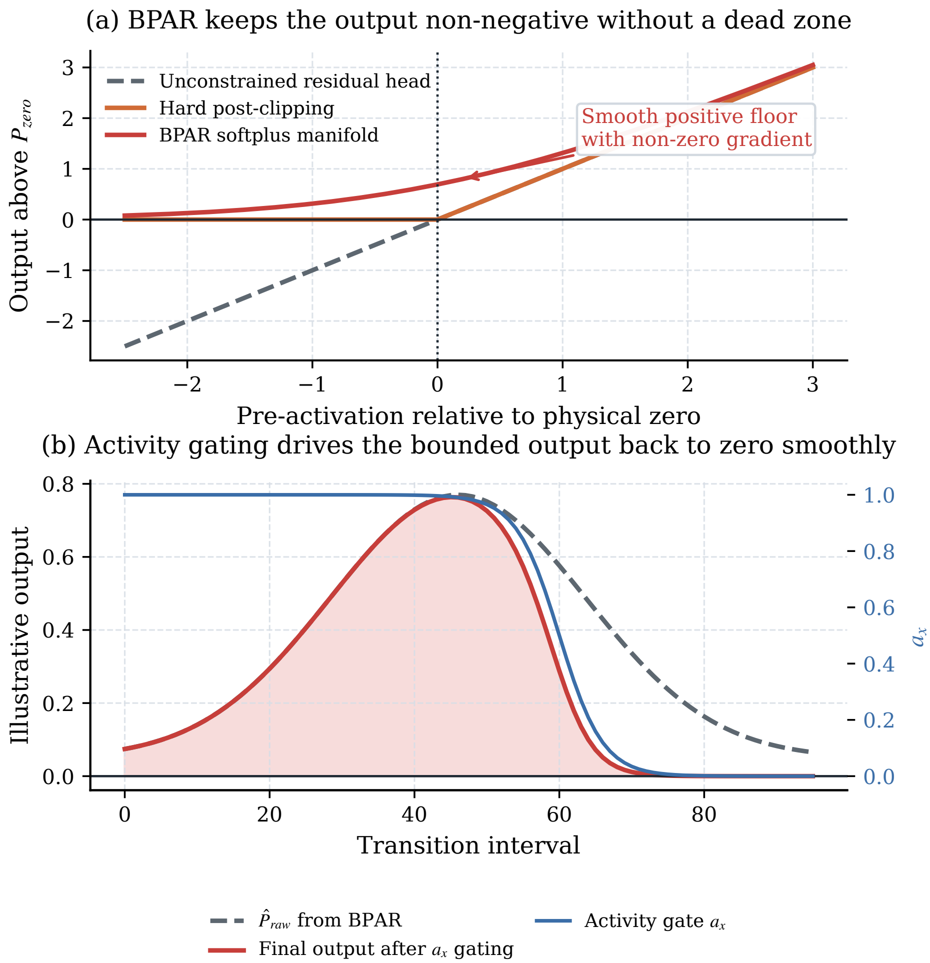

Figure 3: BPAR removes the hard dead zone of post-clipping and, together with activity gating, smoothly pulls the forecast back toward the physical zero boundary. Panel (b) is an illustrative mechanism plot under the adopted normalization setting.

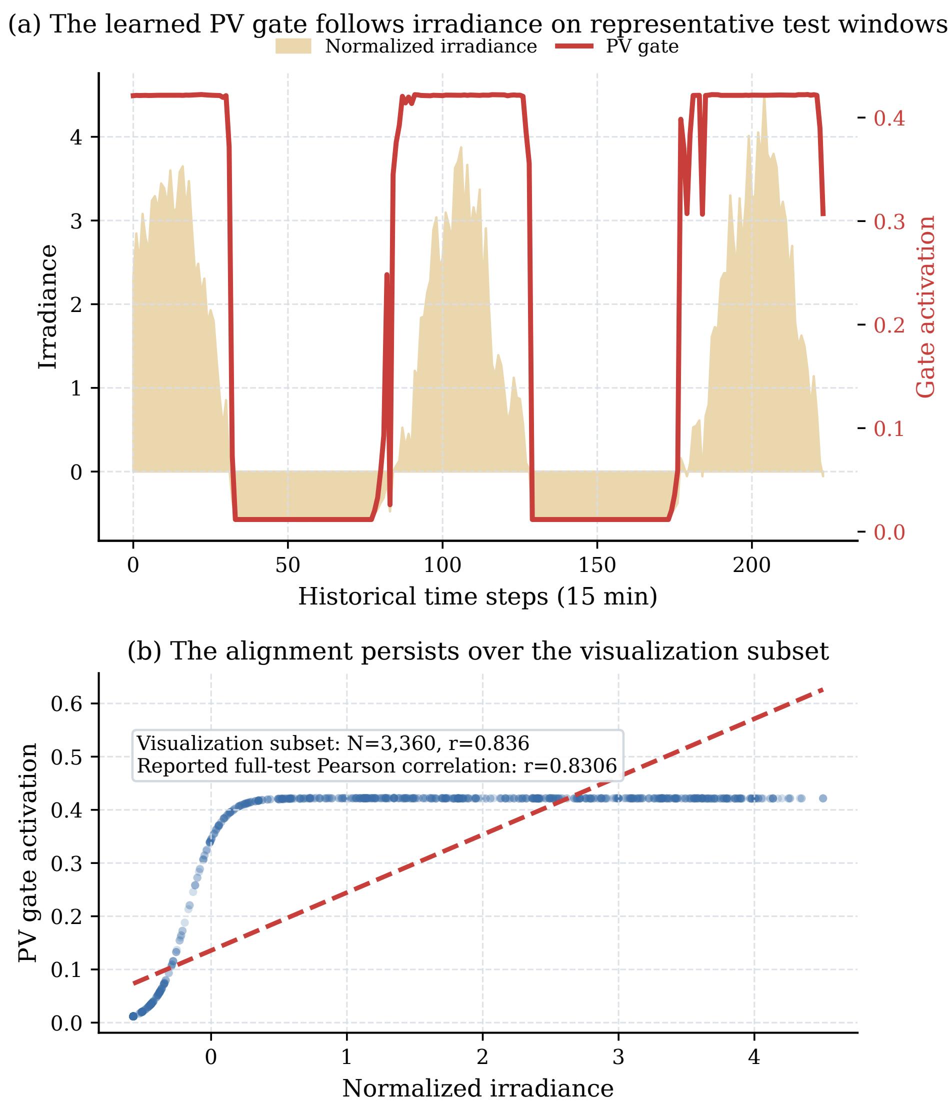

Figure 4: PGCC keeps the learned PV gate aligned with irradiance over both representative windows and the saved visualization subset. The full-test Pearson correlation reported in the text remains $r = 0.8306$.

2.7. Physics-Constrained Curriculum Training

Training with physical constraints introduces conflicts between the forecasting loss and the physics losses. We therefore use a Physics-Constrained Curriculum Training strategy. It separates metric spaces, balances amplitudes dynamically, and activates constraints in phases.

2.7.1. Loss Function Design

The regression loss (MSE) is computed in normalized space for stable optimization. All physics constraints (L_{phys}) are computed after denormalization in MW. We use six penalties (Table 1): four soft constraints for structure and two hard constraints for safety-critical violations.

Table 1: Components of Physics-Guided Loss Function

Comp.	**Wt.**	**Formula**	**Motivation**
L_{net} (Soft)	1.0	$\lvert\sum \hat{y}_{\text{comp}} - y_{\text{net}}\rvert$	Ensures models sum to net demand.
L_{energy} (Soft)	0.5	$\lvert\mathbb{E}_t[\hat{y}] - \mathbb{E}_t[y]\rvert$	Enforces macro energy conservation.
L_{deriv} (Soft)	0.3	$\lvert\Delta\hat{y} - \Delta y\rvert$	Captures ramp rate magnitudes.
L_{dir} (Soft)	0.1	$\mathbb{E}[1-\text{sim}_{\cos}]$	Preserves accurate trend switching.
L_{bvr} (Hard)	2.0	$\mathbb{E}[(\max(0, -\hat{y}_{\text{real}}))^2]$	Penalizes negative power strictly.
L_{rvr} (Hard)	2.0	$\mathbb{E}[(\max(0, \lvert\Delta\hat{y}\rvert - \rho_k))^2]$	Penalizes sudden jumps.

We use an adaptive amplitude balancing factor γ_{ema} to track the ratio between forecast error and physical loss:

$$\gamma_{\text{ema}} \leftarrow 0.9 \cdot \gamma_{\text{ema}} + 0.1 \cdot \left(\frac{L_{\text{mae\_phys}}}{L_{\text{phys\_ref}}}\right) \tag{11}$$

where $L_{\text{phys\_ref}}$ is the total unweighted physical loss in MW. This term yields a dimensionless multiplier. It is set to 1.0 for the first 50 batches, frozen when curriculum activation is below 0.05, and clamped to $[0.1, 10.0]$. The combined objective becomes:

$$L_{\text{total}} = L_{\text{mse}}(\text{normalized}) + \gamma_{\text{ema}} \cdot L_{\text{phys\_weighted}}(\text{MW}) \tag{12}$$

Note: while BPAR provides a structural output floor, L_{bvr} remains essential during training to provide gradient signals that prevent latent representations from drifting into non-physical regimes.

2.7.2. Four-Phase Curriculum

To stabilize optimization, we use the four-phase constraint schedule listed in Table 2.

Table 2: Four-Phase Physics Curriculum and Layer-wise Learning Rates

Phase	**Ep.**	**Active Const.**	**Notes**
I (Init.)	0–4	None	Physics frozen; base mapping.
II (Safety)	5–14	$L_{\text{net}}, L_{\text{bvr}}, L_{\text{rvr}}, L_{\text{dir}}$	Safety bounds ramped.
III (Integ.)	15–29	Phase II + $L_{\text{energy}}, L_{\text{deriv}}$	Macro conservation.
IV (Satur.)	30–99	All	Full regularized training.
Layer-wise Learning Rates			
Physics (ExplicitPhysicalMapping)		10^{-5}	Preserves priors.
Gating (PGCC)		5×10^{-5}	Stable alignment.
Statistical (all other)		10^{-4}	Fast convergence.

The curriculum introduces the constraints progressively. Phase I learns the statistical backbone with frozen physics parameters. Phase II focuses on safety boundaries. Phase III adds macro-level structure. Phase IV activates the full objective. Each constraint weight is ramped with a bounded sigmoid. The PGCC alignment coefficient follows $\alpha = 1.0 - w_{\text{curr\_net}}$.

2.7.3. Layer-wise Learning Rates

We further separate the optimizer into three AdamW parameter groups (Table 2, bottom). The statistical backbone uses a learning rate of 1×10^{-4}. After epoch 5, the physical parameters are unfrozen at 1×10^{-5}, and the PGCC parameters use 5×10^{-5}.

2.7.4. Regularization and Early Stopping

We use three regularization paths. First, a Physics Prior Regularizer compares the learned physical parameters with nameplate values by mean squared error (weight 0.1). Second, a Gate Response Regularization anchors the temporal PV gate to irradiance (weight 0.05), with dynamic masking when the prior variance $\sigma^2_{\text{prior}} < 10^{-3}$ to avoid night-time noise.

Role of PGCC: In this framework, PGCC mainly improves gate alignment and interpretability rather than raw forecasting accuracy. The final prediction is still produced by the residual pathway, so the alignment term can shape weather-response behavior without dominating the forecasting loss.

Gradient clipping uses a dynamic bound of 0.5 during critical curriculum transitions ($0.1 < w_{\text{curr\_net}} < 0.9$), releasing to 1.0 otherwise. Early stopping uses channel-equalizing $NRMSE_{\text{avg}}$ with patience of 15 epochs.

3. Case Study and Performance Evaluation

3.1. Dataset and Experimental Setup

We evaluate PhysFormer on a real VPP dataset built from aggregated regional measurements of a distributed energy cluster in Germany. The dataset contains 15-minute load, PV, and wind trajectories over three years (2012–2014), for a total of 105,120 samples. We use a chronological split of 70% for training, 10% for validation, and 20% for testing. All inputs are standardized with global Z-score parameters (μ, σ), which are later used in the BPAR re-projection step.

The main model settings are $d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$, $n_{\text{heads}} = 8$, and $e_{\text{layers}} = 3$. For the efficiency comparison, we benchmark PhysFormer against Informer with the same hidden dimension. PhysFormer has 11.4M parameters, whereas Informer has 17.9M. The reported inference latency of 2.10 ms/sample was measured on an NVIDIA RTX 3090 GPU with batch size 1.

We compare PhysFormer with eight baselines: LSTM, GRU, PINN [17], Informer [6], Autoformer [7], PatchTST [8], DLinear [9], and iTransformer [21]. To keep the comparison fair, all models receive the same future weather covariates $\mathcal{X}_{\text{weather\_future}} \in \mathbb{R}^{B \times P \times 3}$. For Transformer baselines, these covariates are added to the decoder inputs. For recurrent baselines, they are used as dynamic covariates.

We report standard accuracy metrics, including mean squared error (MSE), mean absolute error (MAE), and root mean squared error (RMSE). We also report three physical-compliance metrics: Boundary Violation Rate (BVR), Mean Violation Severity (MVS), and Ramp Violation Metric (RVM). BVR measures how often the prediction violates known physical limits, and MVS measures the average magnitude of those violations. For temporal consistency, we use the Ramp Violation Rate (RVR). We also report the Net Mean Absolute Error (NET MAE), defined as mean($|\sum_c \hat{y}_c - y_{\text{net}}|$). The ramp threshold ρ_k is set from the 99.9th percentile of training-set differences, and the strict version (RVR-S) uses the 99.5th percentile without a safety factor.

3.2. Forecasting Performance Analysis

The main results are reported in Table 3. PhysFormer reaches an MSE of 0.0128, close to the strongest unconstrained baselines (Informer at 0.0121 and iTransformer at 0.0122), while reducing BVR to 0.00%. By contrast, the unconstrained baselines achieve slightly lower MSE but still produce non-physical outputs. For example, iTransformer yields a BVR of 16.26%.

Under the strict ramp threshold, PhysFormer records an RVR-S of 0.198%, compared with 0.028% for iTransformer. This gap mainly reflects small local corrections near the zero boundary. The average correction remains on the order of 2.6×10^{-5} MW.

PatchTST and DLinear perform worse in this VPP setting than the stronger multivariate models. This result suggests that cross-channel interaction remains important in aggregated multi-source energy systems. The physics-informed neural network (PINN) baseline benefits from physical inductive bias and outperforms LSTM and GRU, but it still trails the strongest Transformer baselines.

Overall, PhysFormer stays close to the accuracy-compliance trade-off frontier. It gives up a small amount of MSE relative to the strongest unconstrained baselines. In return, it maintains zero-bound compliance and

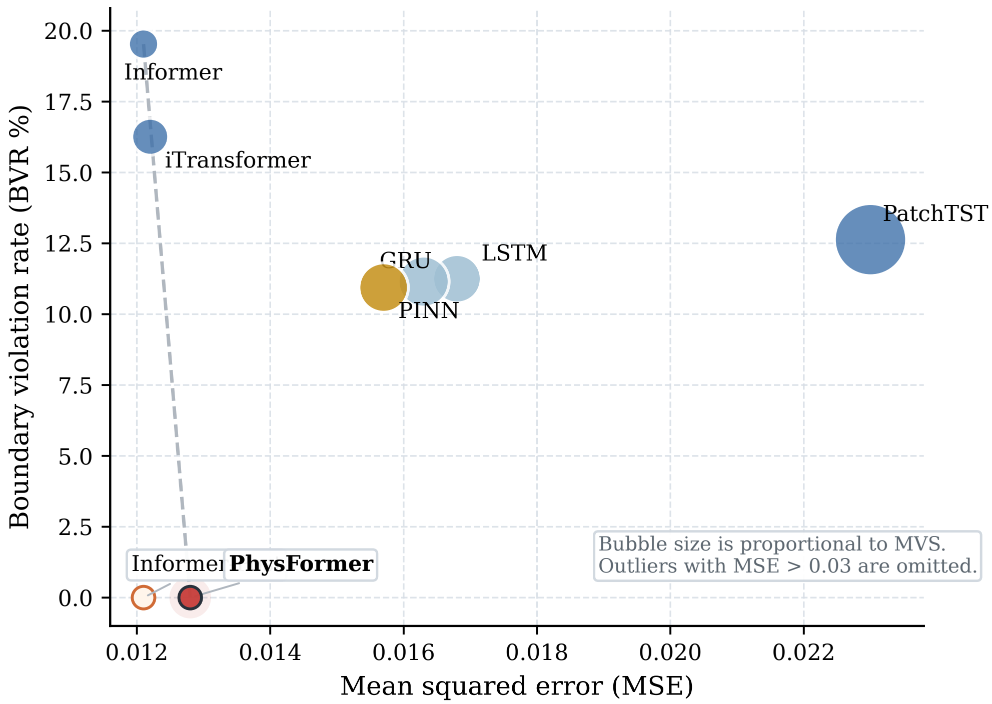

Figure 5: PhysFormer combines zero boundary violations with accuracy close to the strongest unconstrained baselines. High-error outliers with MSE > 0.03 are omitted.

weather-aligned gating behavior, as shown in Fig. 5.

To make the trade-off concrete, Fig. 6 focuses on a representative PV forecast window where the shutdown transition is visible within the 96-step horizon. The PV channel is shown here because boundary handling is most transparent in the zero-generation intervals.

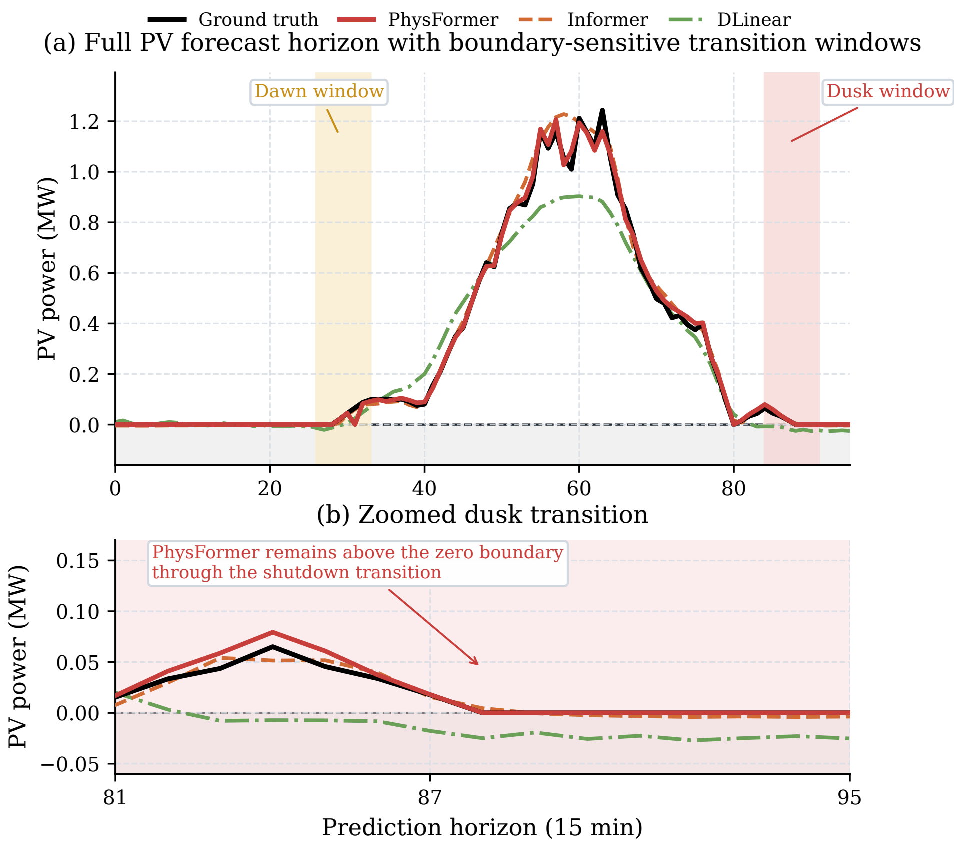

Figure 6: PhysFormer tracks the PV shutdown transition without crossing the zero boundary. Panel (a) shows the full forecast horizon, and panel (b) zooms into the dusk shutdown window.

Table 3: Forecasting Performance on the Global Test Set

Model	MSE	BVR (%)	MVS (MW)	RVR-S (%)	NET MAE
LSTM	0.0168	11.25%	0.0093	0.01%	0.1760
GRU	0.0163	11.15%	0.0103	0.01%	0.1740
PINN	0.0157	10.94%	0.0097	0.02%	0.1702
Informer	**0.0121**	19.53%	0.0036	0.05%	**0.1471**
Informer-Post	0.0121	0.00%	0.0000	0.05%	0.1471
iTransformer	0.0122	16.26%	0.0054	0.028%	0.1482
Autoformer	0.1253	12.67%	0.0737	**0.00%**	0.4846
DLinear	0.1015	7.03%	0.0295	**0.00%**	0.4279
PatchTST	0.0230	12.63%	0.0207	**0.00%**	0.2052
PhysFormer	0.0128	**0.00%**	**0.0000**	0.198%	0.1527

3.3. Verification of Physical Compliance and Safety

3.3.1. Limitations of Heuristic Post-Processing

Table 3 also shows that hard zero-clipping applied to Informer (Informer-Post) reduces BVR to zero without changing the reported MSE. This makes Informer-Post a useful practical baseline rather than a strawman. The difference is structural. Post-processing acts only after prediction, whereas BPAR changes the output parameterization during training and keeps the boundary treatment differentiable.

3.3.2. Robustness Under Extreme Conditions

To evaluate model stability under stronger distribution shifts, we isolate the top 10% highest-volatility samples. The corresponding results are listed in Table 4.

Under these conditions, channel-independent architectures such as Autoformer and DLinear degrade more strongly, whereas the competitive tier remains more stable. PhysFormer keeps BVR at 0.00% while remaining close to the best MSE values. We also checked BPAR under seasonal shifts and degenerate variance conditions ($\sigma_x < 10^{-3}$). For the aggregated VPP windows used here (seq_len = 672), such cases do not appear in the test set.

Table 4: Extreme Weather Performance (Top 10% Volatility)

Model	MSE	BVR (%)	MVS (MW)	RVR-S (%)	NET MAE
LSTM	0.0192	13.43%	0.0086	0.000%	0.1941
GRU	0.0182	13.79%	0.0097	0.000%	0.1864
PINN	0.0177	13.29%	0.0082	0.000%	0.1807
Informer	**0.0146**	21.38%	0.0017	0.070%	**0.1632**
iTransformer	0.0143	19.79%	0.0053	**0.046%**	0.1639
PatchTST	0.0261	12.22%	0.0132	0.000%	0.2240
PhysFormer	0.0150	**0.00%**	**0.0000**	0.294%	0.1715

Fig. 7 shows the same pattern: PhysFormer remains close to the best MSE tier while keeping zero BVR.

3.4. Model Interpretability and Efficiency

3.4.1. Evaluation of Physical Interpretability

A key advantage of PhysFormer is that its physical parameters remain readable after training. As shown in Table 5, the learned values stay within typical engineering ranges [18, 19]. The PV temperature coefficient changes by only +2.5%, and the wind thresholds stay close to their initial priors.

The main quantitative evidence is the alignment between the learned PV gate and irradiance. PGCC yields a global Pearson correlation of $r =$

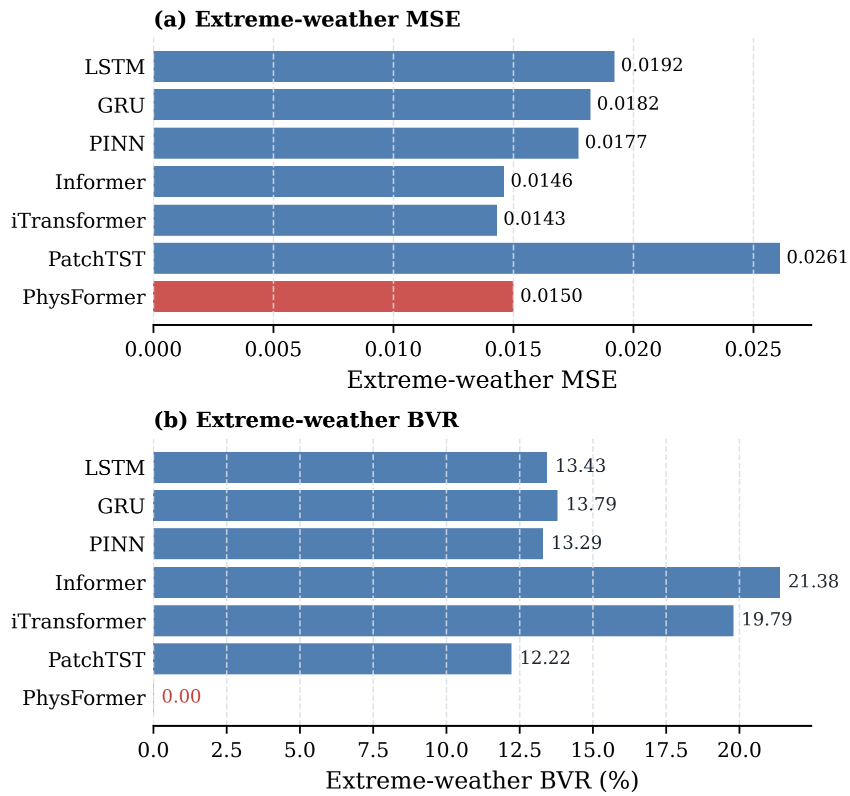

Figure 7: On the extreme-weather subset, PhysFormer stays near the best MSE tier and is the only model with zero boundary violations.

Table 5: Physical Parameter Convergence Tracking

Parameter	Initial	Converged	Lit. Range	Δ
`pv_temp_coef`	0.0040	0.0041/°C	0.003–0.005	+2.5%
`wind_cut_in`	3.50 m/s	3.458 m/s	3–5 m/s	−1.2%
`wind_rated`	12.00	11.970 m/s	10–15 m/s	−0.2%
`wind_cut_out`	25.00	24.968 m/s	20–25 m/s	−0.1%
`load_base`	2.832 MW	2.823 MW	≈ mean	−0.3%
`temp_comfort`	20.0°C	19.992°C	18–22°C	≈ 0

0.8306 ($p < 0.001$). On the daytime subset (irradiance > 0.1), the correlation remains significant at $r = 0.4058$ ($p < 10^{-57}$).

This behavior should be understood as supervised alignment rather than emergent causality. Under additional sensor-noise tests, the correlation remains at $r = 0.8377$ with 20% white noise and stays above 0.81 even when the noise amplitude reaches 50%.

3.4.2. Distinction Between Supervised Alignment and Emergent Causality

It is important to state the limits of this result clearly. The high correlation ($r = 0.8306$) does not indicate emergent meteorological causality discovered *ab initio* from raw data. Instead, Gate Response Regularization provides supervised alignment to a known physical prior.

3.4.3. Computational Complexity Analysis

To evaluate computational cost, we compare parameter counts and inference latency under the same hardware setting. As shown in Table 6, PhysFormer uses $d_{\text{model}} = 512$ but has 11.4M parameters, compared with Informer's 17.9M. The measured latency is about 2.10 ms/sample on an RTX 3090 GPU.

Table 6: Model Complexity and Inference Overhead

Model	Params (M)	Latency (ms)
LSTM / GRU	5.42 / 4.10	1.20 / 2.19
PINN / DLinear	2.74 / 0.13	<0.10 / <0.10
Informer / Autoformer	17.90 / 17.88	1.96 / 4.28
PatchTST / iTransformer	1.62 / 9.85	0.10 / 1.81
PhysFormer	11.4	2.10

3.5. Ablation Study

We further evaluate the contribution of each module through an ablation study on the real dataset (Table 7).

3.5.1. Component-wise Ablation Analysis

Table 7 shows four main trends. First, removing the physics stream increases MSE by 4.7%, so the theory-based baseline remains useful. Second, removing PGCC leaves MSE unchanged, which is consistent with its role in alignment rather than raw accuracy. Third, removing curriculum training increases MSE by 8.7%, which indicates that staged optimization helps. Fourth, removing the future GLU increases MSE by 1.6% while keeping BVR at 0.00%, which suggests that future-weather injection mainly improves accuracy around transition periods.

An additional variant ("w/ PGCC but w/o Gate Response Regularization") keeps a competitive MSE (0.0128) but fails to produce clear gate-weather alignment ($r < 0.12$). This confirms that explicit regularization is the main source of the alignment result.

The evaluation also has clear limits. All experiments are conducted on one aggregated regional dataset with one temporal resolution and one forecasting

Table 7: Ablation Study on Real Data

Variant	MSE	BVR (%)	RVM	$r(\text{gate}_{\text{pv}})$
Full PhysFormer	0.0128	0.00	0.0000	0.8306
w/o Physics Stream	0.0134	0.00	0.0000	N/A
w/o PGCC	0.0128	0.00	0.0000	N/A
w/o Future GLU	0.0130	0.00	0.0000	0.8024
w/o Curriculum	0.0139	0.00	0.0000	0.7866

horizon. The study focuses on component-level forecasting and does not include storage dynamics, state-of-charge constraints, or closed-loop dispatch. The reported interpretability result should therefore be read as supervised alignment to known weather cues, not as causal discovery.

4. Conclusion

This paper presents PhysFormer, a physics-guided Transformer for physically consistent VPP forecasting. The model combines an explicit physical mapping stream, a bounded residual output mechanism, a physics-guided gating and alignment module, and curriculum-based constraint training.

On the 3-year VPP dataset used in this study, PhysFormer achieves a boundary violation rate of 0.00% and remains competitive with strong unconstrained Transformer baselines in mean squared error. The results suggest that the proposed design offers a useful balance between forecasting accuracy, physical consistency, and interpretable weather-response behavior. PGCC mainly improves alignment and interpretability, while BPAR is responsible for the consistent non-negative output behavior.

This study is limited to one regional dataset, one prediction setting, and component-level forecasting without explicit storage, state-of-charge, or dispatch feedback. The interpretability results should also be understood as supervised alignment to known physical cues rather than causal discovery. Future work will extend the framework to additional datasets and forecasting horizons. It will also examine how physically consistent forecasts can support downstream VPP scheduling and control [1, 16, 11].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, manuscript restructuring, and formatting adjustments. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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